Weight Quantization and the Reliability of Turkish Linguistic Measurements in EuroLLM

Abstract

Computational research on Turkish texts requires transparent evidence about the stability of its linguistic instruments. This study audits how weight quantization changes grammatical preferences in EuroLLM-1.7B, treating minimal-pair performance as a bounded diagnostic rather than a validated measure of social meaning. The local FP16 reference is compared with MLX affine 8-bit and 4-bit configurations on 1,000 deduplicated TurBLiMP pairs across four categories. Mean token log probability defines the primary score. FP16, Q8, and Q4 achieve accuracies of 91.2%, 91.5%, and 88.4%. The Q8 difference of +0.3 percentage points is not significant and does not establish equivalence. Q4 loses 2.8 percentage points (Holm-adjusted p = .000469), with 42 correct-to-incorrect and 14 incorrect-to-correct transitions. Subject agreement accounts for 18 of the 28 net lost decisions. Summed scoring increases the Q4 deficit to 3.9 percentage points, showing sensitivity to operationalization. Exploratory analyses do not support monotonically increasing reversals with subword fragmentation. Measured peak MLX allocator memory is 63.8% lower for Q4. A measurement-oriented audit links these findings to requirements for validating local text-analysis instruments: explicit constructs, paired comparisons, disaggregated errors, label review, and task-specific validation. The experiment establishes a model-specific accuracy–memory trade-off; it does not establish the validity of downstream social-science coding, equitable access, or equivalence across precision settings.

Keywords: weight quantization, Turkish, measurement validity, computational social science, morphosyntax, reproducibility.

1. Introduction

When a language model is used to turn texts into research observations, the stability of that instrument matters alongside its computational cost. A model may help identify linguistic patterns without supplying valid measurements of the social concepts a researcher intends to study. Grimmer and Stewart (2013) emphasize that automated text analysis requires extensive, problem-specific validation. This concern motivates the present question: what evidence should researchers require before treating a memory-reduced language model as a dependable linguistic component of a Turkish text-analysis workflow?

The empirical contribution is an audit of grammatical preference under changes in weight precision. It addresses one part of instrument evaluation: whether the same fixed texts receive the same linguistic decisions, how those decisions relate to benchmark labels, and how the answer depends on scoring and

sample construction. This is relevant to the preparation of computational research instruments, but it is not a direct evaluation of sentiment, political stance, responsibility attribution, or any other social-science variable.

Running language models on local hardware requires sufficient memory for both the weights and the computations performed during inference. Representing weights at lower precision reduces their storage requirements. Whether the resulting model preserves the grammatical distinctions made by its reference counterpart is a separate empirical question. Multilingual evaluations have shown that the effects of quantization vary across languages and tasks and that automatic metrics can underestimate the degradation observed by human evaluators (Marchisio et al., 2024).

In Turkish, grammatical relations such as agreement, case selection, and subordination are often expressed through morphology. A single word can contain several morphemes, while a language model's tokenizer can place boundaries that do not coincide with morpheme boundaries. Sentence length and the distribution of morphemes can therefore affect the interpretation of minimal-pair scores (Başar & Bisazza, 2026). A decline in accuracy alone does not establish which grammatical distinction has become harder to recognize.

We evaluate three precision configurations of EuroLLM-1.7B (Martins et al., 2024) on the same Turkish minimal pairs from TurBLiMP (Başar et al., 2025). Three empirical questions organize the audit. RQ1 asks how 8-bit and 4-bit quantization change benchmark accuracy and individual decisions relative to FP16. RQ2 asks whether the conclusions are stable across grammatical categories, sentence-scoring conventions, and sensitivity analyses for related or disputed pairs. RQ3 asks what accuracy–memory trade-off is observed under the measured local workload. An exploratory analysis examines associations between subword fragmentation and Q4 reversals. These questions delimit the evidence that this experiment can contribute to evaluating a linguistic instrument.

The methodological contribution is a reporting procedure that keeps numerical stability, benchmark agreement, and application validity separate. This distinction prevents a small aggregate accuracy difference from being interpreted as evidence that a compressed model can replace its reference in any Turkish research task. The discussion translates the audit into requirements for a subsequent study of actual research coding; it does not present that subsequent study as completed.

2. Related Work

2.1. Low-precision Inference

Quantization schemes with the same nominal bit width can implement different numerical operations. LLM.int8() retains a higher-precision computation path for outlier features while using 8-bit matrix multiplication for most values (Dettmers et al., 2022). GPTQ uses approximate second-order information for post-training weight quantization (Frantar et al., 2023). AWQ uses activation statistics

to identify scaling factors that protect salient weights (Lin et al., 2024). SmoothQuant redistributes quantization difficulty between activations and weights to support 8-bit computation for both (Xiao et al., 2023). The present comparison holds the MLX affine scheme and group size fixed and varies the weight bit width. No further training or task-specific fine-tuning was performed.

2.2. Multilingual and Turkish Evaluations

Marchisio et al. found that quantization affected languages differently across multilingual models (Marchisio et al., 2024). Their results motivate language-specific evaluation but do not establish a ranking of Turkish relative to other languages. Bayram et al. examined quantization in Turkish classification and summarization tasks alongside system efficiency (Bayram et al., 2025). Our outcome differs from those task scores: we track whether a model changes its preference between two sentences and whether each change agrees with the benchmark label. Differences in tasks and scoring preclude a direct ranking across the studies.

2.3. Minimal Pairs and Morphology

BLiMP uses minimally contrasting sentences to probe language models' sensitivity to grammatical distinctions (Warstadt et al., 2020). TurBLiMP extends this approach to 16 Turkish linguistic phenomena, including constructions involving flexible word order and morphological subordination (Başar et al., 2025). Başar and Bisazza investigated how morpheme counts, subword counts, length, and tokenizer alignment with morphological boundaries affect such evaluations (Başar & Bisazza, 2026). We treat minimal-pair accuracy as a measure of sensitivity to the selected contrasts. Our token-to-word ratio is a sentence-level proxy for fragmentation and does not measure morpheme counts.

2.4. Validation in computational text research

Adcock and Collier (2001) distinguish the conceptual and observational levels involved in measurement and emphasize the contextual specificity of validity claims. This provides a useful starting point for assessing language-model outputs: the meaning of a target concept must be specified before a score can be interpreted as its indicator. For this study, the operational target is preference for the benchmark-designated grammatical alternative, not general communicative competence or comprehension of a social claim.

Gilardi et al. (2023) evaluate ChatGPT on annotation tasks involving tweets and news articles, including relevance, stance, topics, and frames. Their study illustrates an application-level evaluation in which model labels are assessed against human annotations. The present task uses sentence likelihoods from a smaller base model and a grammatical benchmark. Findings from either setting cannot establish validity in the other without additional evidence. A successful linguistic audit is therefore one source of diagnostic information, rather than a substitute for a validated coding scheme.

3. Measurement Framework

3.1. Weight Approximation and Preference Margins

Groupwise affine quantization represents weights approximately using low-bit integer codes and a scale and offset shared within each group (MLX Contributors, n.d.). Approximation error can alter the probabilities assigned to sentence tokens without changing the preferred sentence. Conversely, a small shift can reverse a preference when the original scores are close. For each pair, the margin is the score of the benchmark-preferred sentence minus the score of its alternative. A positive margin indicates agreement with the benchmark. A change of sign indicates a reversal.

An association between reversals and small absolute baseline margins is partly a consequence of this decision rule. A point close to zero requires a smaller displacement to cross the boundary. Such an association can identify vulnerable decisions without identifying the representation of a particular suffix as the cause.

3.2. Tokenization and Morphosyntax

With a fixed tokenizer, changing weight precision leaves the token sequence of a given sentence unchanged. Matching token identities across conditions is therefore a validity check. The empirical question is whether an existing characteristic of those sequences helps explain which decisions change. A suffix may be represented by different subword units in different contexts. The available records permit a test using a fragmentation proxy but do not identify morpheme boundaries or changes in internal representations.

3.3. Levels of evidence and permissible inference

We organize the audit around three distinct relationships. First, configuration stability concerns whether changing weight precision changes the decision for a fixed pair. Second, benchmark agreement concerns whether that decision matches the supplied label. Third, application validity concerns whether an output measures the construct of interest in the intended corpus. The first relationship can be evaluated without assuming that every benchmark label is correct. The second depends on label quality. The third requires evidence that is absent from the present grammaticality experiment.

This distinction matters because identical errors can produce perfect stability. Conversely, a reversal can improve agreement with the benchmark. We therefore report both directions of change and retain the underlying continuous margins. A model that reproduces the reference is not necessarily a valid instrument, and a model that differs from it is not necessarily worse on every item. These statements follow from the definitions of stability and accuracy; they do not assume that FP16 provides ground-truth language judgments.

The audit also distinguishes the measurement rule from the computational configuration. The rule specifies how token probabilities become a sentence score and how two scores become a binary decision. The configuration specifies the model weights, tokenizer, numerical representation, and runtime. Holding one fixed while varying the other makes their contributions observable. When both scoring conventions and weight precision change, their effects should not be collapsed into an unspecified change in model quality.

A possible application is the coding of who performs an action in a news sentence. Person agreement and argument structure may be relevant to interpreting that sentence, but grammatical preference does not itself identify an actor or establish responsibility attribution. Connecting the diagnostic to that application would require an explicit actor-coding definition, a representative corpus, independent human judgments, and direct evaluation of the resulting codes. This example motivates a validation requirement; no actor-coding outcomes are inferred from the benchmark.

4. Data and Task

We used four TurBLiMP categories: subject agreement, transitive argument structure, nominalization, and irregular forms. Each source file contained 1,000 pairs. We screened for empty sentences, identical alternatives, and exact duplicate pairs. No invalid rows or cross-category exact duplicates were found. Removing 338 within-category duplicates left a pool of 3,662 distinct pairs. Using Python's random.Random(42) and a fixed category order, we sampled 250 pairs from each category. See Table 1.

Table 1

Sampling Frame and Selected Pairs.

Source: Author's calculations from the experimental records.

Category	Raw	Duplicates	Unique	Selected
Subject agreement	1000	66	934	250
Argument structure	1000	165	835	250
Nominalization	1000	55	945	250
Irregular forms	1000	52	948	250
Total	4000	338	3662	1000

A duplicate required an exact match of both sentences within a category.

The source revision recorded in the project manifest was 297de13fb7a0ce524fe32e8b175c6b5255d66960. The SHA-256 digest of the selected JSONL sample was bcb2c7f25475f8290d1511d742e1e9f8f515fd29df11dd3e3f64f2f53bcb4835. Resampling the local

source files reproduced the content and order of the selected pairs. The digest documents file integrity. Linguistic label validity requires separate evidence.

The sample contained 1,000 unique pair identifiers and 1,000 distinct sentences on each side of the contrast. Equal category sizes make micro-accuracy identical to the unweighted mean of the four category accuracies. These weights describe the benchmark design rather than the prevalence of the phenomena in naturally occurring Turkish. Ellipsis, morphological tagging, and inflected-word generation were outside the evaluation scope.

Each sentence was scored separately. A decision was counted as correct when the benchmark-preferred sentence received the higher score. The task required neither free generation nor a verbal grammaticality judgment. The preferred and dispreferred sentences averaged 13.415 and 13.464 tokens, respectively, with ranges of 6–28 and 6–29. No truncation was required within the 4,096-token context window. Each configuration scored 26,879 sentence tokens, excluding the beginning-of-sequence context.

5. Methods

5.1. Model Configurations

EuroLLM-1.7B is a multilingual decoder-only base model that includes Turkish (Martins et al., 2024; UTTER Project, n.d.). The local configurations shared the same architecture and tokenizer. The model has 24 layers, a hidden dimension of 2,048, 16 attention heads, and eight key/value heads. Q8 and Q4 used affine weight quantization with a group size of 64. No chat template, conversation history, demonstrations, or task-specific adaptation was used.

The reference is designated FP16 on the basis of the F16 tensors in the local weight file. The BF16 field in the source configuration describes a different aspect of the model. The quantized files contain packed U32 weights and F16 auxiliary tensors. The Q4 designation therefore refers to weight representation and does not imply that all arithmetic was performed at four bits. Project records associate all configurations with the same base model. An incomplete archive of the exact upstream revision, conversion commands, and tensor inventory limits independent reconstruction.

5.2. Sentence Scoring

Sentences were tokenized without automatically adding special tokens. One BOS token supplied the initial context. No EOS token was appended or scored. Each observed sentence token was scored conditional on the preceding context. Logits were cast to float32 before log-softmax, and the log probabilities of the observed tokens were averaged. This cast applied to score calculation rather than the entire model.

$$S_q(s) = \frac{1}{T}\sum_{t=1}^{T} \ln p_q(x_t \mid BOS, x_{<t}) \quad (1)$$

Here T is sentence length in tokens and q denotes the configuration. For pair i, the preferred and dispreferred sentences are denoted by s^+ and s^-. Accuracy is the proportion of strictly positive margins.

$$m_{q,i} = S_q(s^+) - S_q(s^-), A_q = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}(m_{q,i} > 0) \quad (2)$$

No exact ties occurred. Summed log probability was retained as a sensitivity analysis. Scores cover complete sentences, so a reversal cannot be attributed specifically to the probability of the critical suffix.

5.3. Paired Inference

The statistical unit was the minimal pair. We compared each quantized configuration with FP16 using a two-sided exact McNemar test (McNemar, 1947). Let d count correct-to-incorrect transitions and u count incorrect-to-correct transitions. Under the null hypothesis, conditional on d + u discordant pairs, the number of transitions in one direction follows a binomial distribution with probability 0.5. We applied Holm adjustment (Holm, 1979) to the two primary comparisons at a significance level of .05. No time-stamped preregistration was documented.

Individual accuracy intervals used the 95% Clopper–Pearson procedure (Clopper & Pearson, 1934). For the paired difference, we calculated separate 97.5% Clopper–Pearson intervals for u/N and d/N. Subtracting the upper bound for degradation from the lower bound for improvement, and vice versa for the opposite endpoint, gives a conservative interval with at least 95% coverage under the independent-pair sampling model. These intervals apply to individual contrasts. Holm adjustment was applied to p values.

5.4. Dependence Among Related Pairs

Distinct pairs may share a linguistic core. Within each category, we grouped pairs whose normalized alternatives matched after removing shared leading words from a specified list: ama, ancak, ardından, artık, aslında, ayrıca, belki, bugün, burada, evet, fakat, hadi, hemen, işte, sadece, sonra, ve, yalnız, yani, yine, çünkü, öncelikle, özellikle, and şimdi. The resulting 776 clusters approximate related examples. Their membership has not been validated against the original generation templates.

We used category-stratified bootstrap resampling (Efron, 1979) of complete pairs and, separately, of complete approximate clusters. Each procedure used 20,000 replicates for the overall accuracy contrast, with seeds 20261028 and 20261029, respectively. Categories retained equal weights. Dependence beyond that captured by the clustering rule may remain.

5.5. Exploratory Fragmentation Analysis

For each pair, the fragmentation proxy was the arithmetic mean of the two token-to-word ratios.

$$F_i = \frac{1}{2}\left(\frac{T^+}{W^+} + \frac{T^-}{W^-}\right) (3)$$

T and W denote token and word counts. Words were counted as Unicode letter or digit sequences, preserving internal apostrophes and treating hyphens as boundaries. Standalone punctuation contributed to the token count but not the word count. The outcome of the logistic regression was any reversal between FP16 and Q4, regardless of direction.

The unadjusted model contained fragmentation alone. The main adjusted model also included absolute FP16 margin, mean word count, and category, with argument structure as the reference. All predictors, including category indicators, were standardized. We added a ridge penalty equal to λ/2 times the sum of squared slopes to the total negative log likelihood and left the intercept unpenalized. The main penalty was λ = 1, with 0.1 and 10 examined for sensitivity. No predictive tuning or held-out classifier evaluation was performed.

Pair and cluster resampling each used 2,000 replicates, with seeds 20261029 and 20261030. We refitted the scales and coefficients in each replicate and normalized category weights to their original sizes. Odds ratios were expressed per original-sample standard deviation of fragmentation. Percentile intervals describe uncertainty in the penalized estimates and do not correct regularization bias or multiplicity across exploratory models. Q8 had only three reversals and was not modeled separately.

5.6. Audit procedure and interpretation rules

The audit treats the pair identifier as the join key across precision settings and verifies that the same token sequences were scored. It retains continuous scores, binary decisions, and benchmark labels so that a reader can separate reference agreement from label agreement. Overall accuracy is accompanied by transition counts, category-specific results, and normalization sensitivity. The resource measurements are attached to the exact scoring workload rather than generalized to all deployments.

Interpretation follows the available evidence. A nonsignificant superiority test is not treated as proof of equivalence. A category-specific loss is not treated as evidence of harm to a demographic group. A memory reduction is not treated as evidence of improved access, lower energy consumption, or lower institutional expenditure. These restrictions define the use of the audit in research planning: it identifies decisions requiring further validation and records the limits of the evidence rather than certifying a model for an untested application.

6. Experiments

The experiment ran on an Apple M2 MacBook with 16 GiB of unified memory, macOS 26.4.1, Python 3.12.15, MLX 0.32.4, and MLX-LM 0.32.2. Batch size was one sentence. We processed 20 chunks of 50 pairs and recorded scoring times after loading and warm-up. The additional analyses used NumPy 2.5.4, SciPy 1.18.1, and Matplotlib 3.11.2.

The audit covered 3,000 pair–configuration records and 6,000 sentence records. Checks included identifiers, categories, token counts and digests, and score consistency. The maximum absolute discrepancy between the stored mean and the sum divided by token count was 4.24×10^{-7}. Recomputing the original statistics produced identical results.

Mean token log probability defined the primary analysis. Summed scores, category breakdowns, pilot exclusion, label sensitivity, margin distributions, and fragmentation models provided secondary or exploratory analyses. The pilot included five pairs per category drawn from the main sample. A complete rerun in separate processes reproduced every decision. It did not add independent observations. Background load was uncontrolled during the rerun, limiting performance comparisons.

7. Results

7.1. Overall Accuracy and Transitions

FP16, Q8, and Q4 produced 912, 915, and 884 correct decisions, respectively. Q8 retained every correct FP16 decision and corrected three errors. The small positive difference did not establish superiority. Equivalence or noninferiority would require a prespecified acceptable loss. See Table 2.

Table 2

Primary Accuracy and Paired Contrasts against FP16.

Source: Author’s calculations from the experimental records.

Model	Correct / N	Accuracy % [95% CI]	Δ pp	Paired 95% CI, pp	Holm p
FP16	912/1000	91.2 [89.27, 92.88]	—	—	—
Q8	915/1000	91.5 [89.60, 93.15]	+0.3	[−0.390, 0.971]	.250
Q4	884/1000	88.4 [86.25, 90.32]	−2.8	[−5.151, −0.416]	.000469

pp = percentage points. Accuracy intervals are exact binomial intervals. Difference intervals use the conservative paired procedure.

Q4 reversed 56 decisions (5.6%). Forty-two were degradations and 14 were improvements, yielding a net loss of 28 correct decisions. FP16 and Q4 were both correct for 870 pairs and both incorrect for 74. The error count rose from 88 to 116, a relative increase of 31.8%. All three configurations agreed on 870 correct and 72 incorrect decisions.

For Q4, the category-stratified pair bootstrap interval was [−4.30, −1.40] pp and the approximate-cluster interval was [−4.45, −1.21] pp. For Q8, bootstrap lower bounds of zero reflect the absence of observed degradations: resampling cannot generate a transition type absent from the observed sample. The conservative interval is therefore the primary basis for interpretation.

7.2. Category Differences

Subject agreement accounted for 18 of the 28 net lost decisions (64.3%). Argument structure and nominalization contributed four and seven net losses, respectively, while irregular forms contributed one net gain. See Table 3.

Table 3

Category Accuracy and Q4 Transitions (250 Pairs per Category).

Source: Author's calculations from the experimental records.

Category	FP16 %	Q8 %	Q4 %	Q4 Δ pp	d / u
Subject agreement	88.4	88.8	81.2	−7.2	23 / 5
Argument structure	93.2	93.2	91.6	−1.6	6 / 2
Nominalization	84.0	84.8	81.2	−2.8	13 / 6
Irregular forms	99.2	99.2	99.6	+0.4	0 / 1

d and u denote degradations and improvements relative to FP16.

Across a separate exploratory Holm family of eight category comparisons, the adjusted p value for Q4 subject agreement was .00730. The other comparisons did not reach the .05 threshold. Differences in significance across categories do not establish an interaction. Irregular forms were already near ceiling at 99.2% under FP16, leaving only two errors available for improvement.

7.3. Baseline Margins

Mean absolute margin shifts relative to FP16 were 0.00838 for Q8 and 0.10509 for Q4. Pearson correlations with FP16 margins were .99939 and .90822, respectively. Close agreement across the full distribution can coexist with sign changes near zero. Across successive absolute FP16 margin quartiles, Q4 reversed 46, 8, 1, and 1 decisions. The lowest quartile accounted for 82.1% of reversals. Its reversal rate was 18.4%, compared with 1.33% among the remaining 750 pairs. See Figure 1.

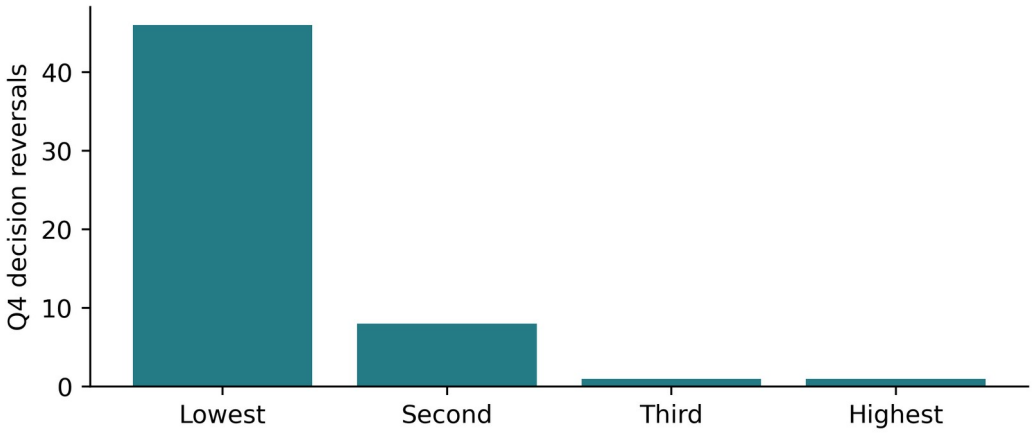


Figure 1

Q4 Reversals by Absolute FP16 Margin Quartile (250 Pairs Each).

Source: Author's calculations from the experimental records.

7.4. Scoring and Sample Sensitivity

Under summed log probability, accuracy was 94.6% for FP16, 94.8% for Q8, and 90.7% for Q4. The Q4 contrast increased in magnitude to −3.9 pp. Switching between mean and summed scoring changed 60 FP16, 59 Q8, and 81 Q4 decisions. In the 626 pairs with equally long token sequences, both scoring rules yielded the same preference for every configuration. See Table 4.

Table 4

Sensitivity of the Q4–FP16 Contrast.

Source: Author's calculations from the experimental records.

Analysis	N	Δ pp	Paired 95% CI, pp	Holm p
Primary mean score	1000	−2.800	[−5.151, −0.416]	.000469
Summed score	1000	−3.900	[−6.176, −1.576]	3.04×10^{-7}
Pilot excluded	980	−2.755	[−5.137, −0.340]	.000711
Eight disputed pairs excluded	992	−2.520	[−4.747, −0.262]	.000940
Predicate screen excluded	983	−2.543	[−4.790, −0.265]	.000940
Subject agreement excluded	750	−1.333	[−3.656, 1.016]	.174317

Holm adjustments apply to the two FP16 comparisons within each analysis.

Excluding the pilot or potentially ambiguous nominalizations preserved the negative Q4 contrast. Excluding subject agreement reduced it to −1.333 pp (adjusted p = .174), illustrating sensitivity to category composition. These exclusions cannot establish label validity in the remaining sample.

7.5. Linguistic Examples

A purposive review covered 20 pairs comprising 12 Q4 degradations, five Q4 improvements, and three Q8 improvements. The review was extended to all 58 unique pairs on which either quantized configuration differed from FP16. The preliminary judgments were AI-assisted. No independent blinded human annotation or inter-annotator agreement was obtained.

For subject_agreement:0532, the preferred sentence was “Ancak o bu projeye çok emek verdi.” The alternative substituted sen (‘you’) for o (‘he/she’) while retaining the third-person predicate verdi. The FP16, Q8, and Q4 margins were 0.301963, 0.304616, and −0.062743. For argument_structure_transitive:0850, “Hadi bu filme bayılıyoruz.” contrasts with “Hadi bu filmi bayılıyoruz.” The verb calls for the dative form filme under the intended reading. The corresponding margins were 0.039436, 0.041425, and −0.167631.

In nominalization:0962, the label favors “Ve şirketin yatırım kararlarını hızla almasına çoğu çalışan şaşırdı.” over the alternative with aldığına. Margins were −0.017708, −0.016909, and 0.236379. Event-oriented and fact-oriented interpretations may license both complements, so Q4’s agreement with the label is an ambiguous linguistic improvement. Of the 58 reviewed pairs, eight raised similar concerns, three were context-sensitive, and five required expert review. The selection does not support a corpus-wide estimate of labeling errors.

7.6. Subword Fragmentation

The sample quartile thresholds were 1.857143, 2.083333, and 2.339286 tokens per word. Ties at a threshold remained in the lower group. The group sizes were 256, 247, 247, and 250. Reversal rates were 5.86%, 4.45%, 8.91%, and 3.20%, with no monotonic increase. The highest-minus-lowest contrast was −2.66 pp (approximate-cluster bootstrap 95% interval [−6.50, 1.41]). See Figure 2.

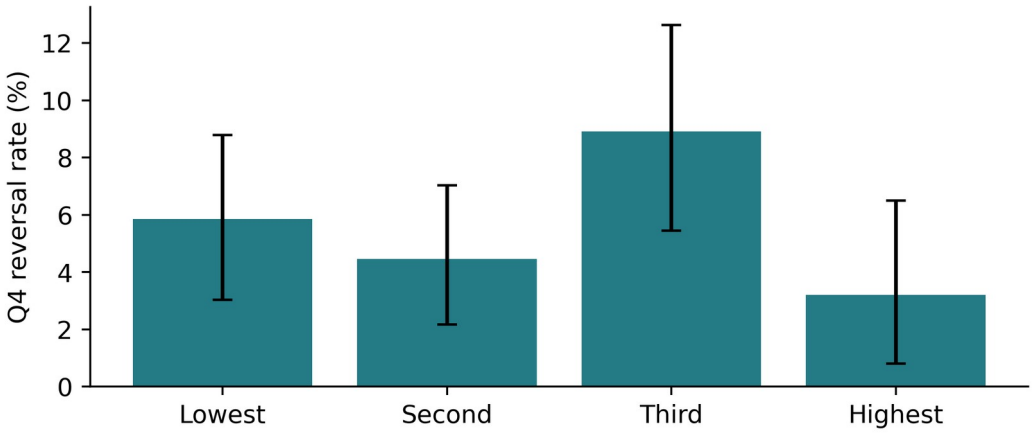


Figure 2

Q4 Reversal Rates by Fragmentation Quartile. Bars Represent 15/256, 11/247, 22/247, and 8/250 Pairs. Error Bars Show Category-stratified Approximate-cluster Bootstrap 95% Intervals.

Source: Author's calculations from the experimental records.

Table 5

Fragmentation and the Odds of a Q4 Reversal.

Source: Author's calculations from the experimental records.

Model	OR per SD	Pair bootstrap 95% CI	Cluster bootstrap 95% CI
Unadjusted	.789	[.611, .982]	[.592, 1.027]
Margin, category, word length	.735	[.548, .932]	[.540, .946]
Margin, category, token length	.900	[.664, 1.193]	[.649, 1.230]

One SD corresponds to 0.3977 tokens per word. Intervals are exploratory and unadjusted for multiple comparisons.

Adjustment for within-pair fragmentation differences yielded an OR of .742 (cluster interval [.544, .952]). Using log absolute margin gave .737 [.546, .953]. Penalties of 0.1 and 10 produced point estimates of .716 and .817, without additional intervals. The negative adjusted association weakened when token count replaced word count. These results offer no evidence for a general protective effect of fragmentation. An odds ratio describes a change in odds, and the outcome includes both improvements and degradations. See Table 5.

7.7. Resource Measurements

In the original run, scoring took 20.4% less time for Q8 and 22.6% less for Q4 relative to FP16. Measured peak MLX allocator memory fell by 63.8% for Q4, and weight-file size fell by 71.9%. See Table 6.

Table 6

Resource Measurements in the Original Run.

Source: Author's calculations from the experimental records.

Measure	FP16	Q8	Q4
Scoring time (s)	123.04	97.88	95.20
Peak MLX allocator memory (GB)	3.416	2.068	1.237
Peak process RSS (GB)	1.247	1.839	1.573
Weight files (GB)	3.314	1.761	0.932

GB denotes 10^9 bytes. Allocator memory, resident set size, and storage have distinct measurement scopes.

Timing includes tokenization, model computation, score extraction, record preparation, and cache clearing between pairs. Loading, warm-up, file writing, and pauses between chunks were excluded. The Q8–Q4 difference of 2.68 seconds represents one run. In the verification run, FP16, Q8, and Q4 took 127.08, 104.86, and 98.85 seconds, respectively, with identical decisions. The 20 chunks contain different sentences and cannot serve as 20 independent performance replicates. Energy consumption, temperature, memory pressure, and swap were not measured concurrently.

8. Discussion

8.1. Stability of the linguistic instrument

The Q4 configuration reduced measured memory use at a cost in accuracy. Q8 remained close to the reference on this sample. The acceptable loss for deployment depends on the intended application, and no practical tolerance was specified before analysis. The observed trade-off consequently applies to the reported workload.

Subject agreement contributed disproportionately to the overall decline. This pattern cannot be attributed specifically to damaged person-suffix representations: lexical choices, template structure, and baseline difficulty vary across categories. No token- or layer-level intervention isolated an internal mechanism. Baseline margins help describe vulnerable examples, although they are not calibrated confidence probabilities or a validated routing criterion.

The fragmentation results complement morphology-aware evaluations (Başar & Bisazza, 2026). A word split into several tokens need not represent a difficult grammatical contrast. Punctuation contributes to the numerator of the proxy, and length enters both the ratio and the adjustment model. The token-length sensitivity interval includes the neutral value, which limits any interpretation of the negative slope as a robust linguistic effect.

Mean and summed scores yielded different absolute accuracy levels and different sizes of the Q4 deficit. Reporting both while retaining the original primary rule makes the effect of normalization visible. Comparisons with other minimal-pair studies should account for their scoring conventions.

8.2. Implications for computational social research

The results expose a practical problem in instrument substitution. A researcher comparing only overall accuracy might regard the observed Q4 decline as a uniform small loss. The transition table and category breakdown show a more selective pattern: some decisions improve, others deteriorate, and subject agreement contributes most of the net loss. If a future coding task relies on particular constructions, its relevant error distribution may differ from the equally weighted benchmark average. This is a reason to validate on the intended task, not evidence that the same pattern will necessarily occur there.

The normalization analysis also locates part of the measurement uncertainty outside the weight representation. On the same sentences, mean and summed log probability yield different accuracy levels and different estimates of the Q4 deficit. Neither normalization can be selected solely because it gives a favorable comparison. A research workflow should specify its scoring rule before the main evaluation, retain an interpretable alternative as a sensitivity check, and explain what construct the resulting score is intended to measure. Here, both rules identify a negative Q4 contrast, while disagreeing about its size.

Benchmark labels introduce a further source of uncertainty. The nominalization examples show why a change toward the preferred label cannot automatically be called a linguistic improvement. Context-dependent alternatives require judgments about intended readings. Reviewing only changed pairs is useful for diagnosing reversals but cannot estimate the prevalence of questionable labels in the full sample. A future validation should therefore combine a probability sample from all categories with a separately identified diagnostic sample of reversals. Those samples answer different questions and should not be pooled without accounting for their selection.

8.3. Resource requirements and the limits of access claims

The measured reduction in allocator memory identifies a computational opportunity under one workload. It does not establish that a particular institution or user can adopt the model, because adoption also depends on available equipment, task requirements, maintenance, expertise, and other

constraints not observed here. Likewise, the experiment does not estimate a monetary benefit or an environmental saving. The appropriate conclusion is narrower: a configuration with substantially lower measured allocator memory produced less reliable agreement with the benchmark, and the distribution of that loss deserves scrutiny before application.

The Q8 result offers a useful comparison but not a deployment recommendation. Its accuracy is close to FP16 in this sample and its weight file and allocator memory are smaller. An equivalence or noninferiority claim would require an application-specific acceptable loss and a suitable design. Choosing such a tolerance after observing the present result would change the inferential question. Researchers can use the reported intervals to plan a future evaluation, but cannot retrospectively convert the superiority test into a prespecified equivalence test.

8.4. A validation workflow for subsequent applications

The proposed workflow begins by defining the social-science construct and its coding unit. Researchers then select texts from the intended setting, develop a coding manual, and obtain independent human labels before comparing model configurations. The development material must be separated from the final evaluation, including any prompt selection or adaptation required for a base model. A precision comparison should keep those decisions fixed. At the final stage, researchers should examine not only individual coding errors but also changes in the aggregate quantity they intend to report, such as the proportion of texts assigned to an actor category.

This workflow connects the current diagnostic to application validity without assuming a numerical transfer from one to the other. For example, a 2.8-percentage-point decline in minimal-pair accuracy does not imply a 2.8-point bias in an estimated responsibility-attribution rate. The relationship depends on the new labels, corpus, and coding procedure. The present contribution is the documented linguistic audit and its interpretation rules; direct validation of a social-science coding task remains an empirical next step.

9. Limitations

We evaluated one base model, one quantization scheme, one language, and four categories. Other model scales, monolingual Turkish models, and calibration-based methods may behave differently. No cross-language control permits attribution to Turkish or to agglutinative languages generally. Overlap between test sentences and pretraining data was not investigated.

Balanced category sampling does not represent natural usage or the entire benchmark. Exact deduplication leaves related templates, and the clustering heuristic captures only selected initial-word variants. Exploratory AI-assisted review does not substitute for blinded judgments by independent Turkish-language experts. Exclusion analyses address only the label issues identified.

Whole-sentence rankings do not measure inflection generation, lemmatization, or morpheme tagging. The fragmentation proxy lacks gold morpheme boundaries. The logistic model estimates a common linear log-odds slope without category interactions or nonlinear terms. Sparse events and regularization limit coefficient interpretation.

The pilot overlaps the main sample, the study was not preregistered, and multiplicity was not controlled across all exploratory analyses. A persistent public experiment archive and complete conversion manifest have not been specified for this version. Decision reproducibility on the same inputs is distinct from independent replication on new items. Timing comparisons were not based on repeated full runs with randomized configuration order.

No social-science corpus was coded and no downstream research estimate was compared across precision settings. The methodological relevance of the audit therefore rests on its role in evaluating a linguistic component before application. It cannot establish the reliability or validity of a completed social-science measurement instrument. The proposed application workflow is a research design recommendation, not a second experiment or an externally validated protocol.

10. Conclusion

Across 1,000 matched Turkish minimal pairs, Q8 differed from FP16 by +0.3 pp without establishing superiority. Q4 differed by −2.8 pp, a statistically significant decline after Holm adjustment. Its 56 reversals included changes in both directions, concentrated in low-margin items. Subject agreement accounted for most of the net decline.

The exploratory analysis did not support an increase in decision changes with greater fragmentation. The inverse conditional association depended on the definition of length. Alongside a 63.8% reduction in measured peak MLX allocator memory, these findings quantify an accuracy–memory trade-off for this model, sample, and scoring procedure. Blinded expert review, verified generation templates, and replication on another checkpoint would test the limits of the result.

For computational research using Turkish texts, the methodological lesson is to document the measurement procedure alongside the model configuration and to evaluate both before substitution. The reported audit makes disagreement, benchmark uncertainty, and resource scope visible. Applying this procedure to an actual social-science coding task is necessary before drawing conclusions about the stability of substantive research findings.

11. Research Transparency

11.1. Data and Code Availability

The original TurBLiMP benchmark is available at https://github.com/ezgibasar/TurBLiMP (Başar et al., 2025). The experiment uses revision 297de13fb7a0ce524fe32e8b175c6b5255d66960 and the

sample digest reported in Section 4. A local replication package accompanies this manuscript preparation and includes the selected data, saved scores, analysis scripts, model configuration files, and an integrity inventory. It does not contain model weights. A persistent public repository link has not yet been assigned. The incomplete archive of model conversion commands limits reconstruction of the weight-conversion stage.

11.2. AI Use Declaration

AI tools assisted with earlier manuscript preparation and preliminary linguistic review, as reported in the experimental records. OpenAI Codex was used during the revision of 6 November 2026 for literature checking, manuscript restructuring, document preparation, and consistency checks against saved results. The exact model identifier for this revision and the tool names and versions used in earlier stages are not documented in the supplied records. AI-assisted linguistic comments are exploratory and are not independent human annotations. The tested EuroLLM configurations are described in Sections 5 and 6. Responsibility for the submitted manuscript, source accuracy, and interpretation remains with the human author.

11.3. Ethics and consent

The reported experiment uses existing benchmark sentences and model-generated likelihood scores. It did not recruit human participants, collect participant responses, or analyze identifiable personal records. No participant consent was collected and no ethics-committee decision is reported. The study is presented as a secondary benchmark analysis; any formal institutional exemption or approval requirement must be determined under the applicable institutional procedure before submission.

References

Adcock, R., & Collier, D. (2001). Measurement validity: A shared standard for qualitative and quantitative research. *American Political Science Review, 95*(3), 529–546. https://doi.org/10.1017/S0003055401003100

Başar, E., & Bisazza, A. (2026). A morphology-aware evaluation of Turkish syntax in large language models. Proceedings of SIGTURK 2026 (pp. 95–102). https://doi.org/10.18653/v1/2026.sigturk-1.9

Başar, E., Padovani, F., Jumelet, J., & Bisazza, A. (2025). TurBLiMP: A Turkish benchmark of linguistic minimal pairs. Proceedings of EMNLP 2025 (pp. 16495–16510). https://doi.org/10.18653/v1/2025.emnlp-main.834

Bayram, C., Turan, C. A., Kürkçüoğlu, F., & Altıntaş, V. (2025). Türkçe LLM'lerde Nicemleme Dengelemesi: Doğruluk, Hız ve Bellek Kullanımı Üzerine Bir Değerlendirme. *Veri Bilimi, 8*(2), 70–80. https://dergipark.org.tr/tr/pub/veri/article/1837255

Clopper, C. J., & Pearson, E. S. (1934). The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika, 26*(4), 404–413. https://doi.org/10.1093/biomet/26.4.404

Dettmers, T., Lewis, M., Belkada, Y., & Zettlemoyer, L. (2022). LLM.int8(): 8-bit matrix multiplication for transformers at scale. Advances in Neural Information Processing Systems, 35, 30318–30332. https://doi.org/10.48550/arXiv.2208.07339

Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics, 7*(1), 1–26. https://doi.org/10.1214/aos/1176344552

Frantar, E., Ashkboos, S., Hoefler, T., & Alistarh, D. (2023). GPTQ: Accurate post-training quantization for generative pre-trained transformers. International Conference on Learning Representations. https://doi.org/10.48550/arXiv.2210.17323

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences, 120*(30), Article e2305016120. https://doi.org/10.1073/pnas.2305016120

Grimmer, J., & Stewart, B. M. (2013). Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis, 21*(3), 267–297. https://doi.org/10.1093/pan/mps028

Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, *6*(2), 65–70. https://www.jstor.org/stable/4615733

Lin, J., Tang, J., Tang, H., Yang, S., Chen, W.-M., Wang, W.-C., Xiao, G., Dang, X., Gan, C., & Han, S. (2024). AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration. Proceedings of Machine Learning and Systems, 6. https://proceedings.mlsys.org/paper_files/paper/2024/hash/42a452cbafa9dd64e9ba4aa95cc1ef21-Abstract-Conference.html

Marchisio, K., Dash, S., Chen, H., Aumiller, D., Üstün, A., Hooker, S., & Ruder, S. (2024). How does quantization affect multilingual LLMs? Findings of the Association for Computational Linguistics: EMNLP 2024 (pp. 15928–15947). https://doi.org/10.18653/v1/2024.findings-emnlp.935

Martins, P. H., Fernandes, P., Alves, J., Guerreiro, N. M., Rei, R., Alves, D. M., Pombal, J., Farajian, A., Faysse, M., Klimaszewski, M., Colombo, P., Haddow, B., de Souza, J. G. C., Birch, A., & Martins, A. F. T. (2024). EuroLLM: Multilingual language models for Europe. arXiv. https://doi.org/10.48550/arXiv.2409.16235

McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika, 12*(2), 153–157. https://doi.org/10.1007/BF02295996

MLX Contributors (n.d.). mlx.core.quantize. MLX documentation (Version 0.32.4). Retrieved November 6, 2026, from https://ml-explore.github.io/mlx/build/html/python/_autosummary/mlx.core.quantize.html

UTTER Project (n.d.). EuroLLM-1.7B [Model card]. Hugging Face. Retrieved November 6, 2026, from https://huggingface.co/utter-project/EuroLLM-1.7B

Warstadt, A., Parrish, A., Liu, H., Mohananey, A., Peng, W., Wang, S.-F., & Bowman, S. R. (2020). BLiMP: The benchmark of linguistic minimal pairs for English. *Transactions of the Association for Computational Linguistics, 8*, 377–392. https://doi.org/10.1162/tacl_a_00321

Xiao, G., Lin, J., Seznec, M., Wu, H., Demouth, J., & Han, S. (2023). SmoothQuant: Accurate and efficient post-training quantization for large language models. Proceedings of Machine Learning Research, 202, 38087–38099. https://proceedings.mlr.press/v202/xiao23c.html